

Untitled

June 29, 2025

1 Sessional - 2

Q1: Answer the following in brief (Any THREE) - [3 Marks Each]

1. What does C, A and P represent in CAP theorem? CAP Theorem, introduced by Eric Brewer, is fundamental in distributed systems. It states that it is impossible for a distributed data store to simultaneously provide more than two out of the following three guarantees: - **Consistency (C):** All nodes see the same data at the same time. This means once data is written, all future reads return that same data. - **Availability (A):** Every request gets a response, without guarantee of the most recent data. - **Partition Tolerance (P):** The system continues to operate despite arbitrary message loss or failure of part of the system.

Most cloud databases (like Cassandra) trade consistency for availability and partition tolerance.

2. Discuss 3 Vs related to Big Data in brief. The 3 Vs of Big Data describe its fundamental characteristics: - **Volume:** Refers to the vast amounts of data generated every second (e.g., social media, IoT sensors). Storage and management require scalable cloud infrastructures. - **Velocity:** The speed at which new data is generated and needs to be processed. Real-time systems like Kafka and Spark are essential. - **Variety:** Data comes in different formats – structured (SQL), semi-structured (XML/JSON), and unstructured (videos, images).

3. Discuss 3 scenarios of portability/interoperability in cloud computing. - **Cloud-to-cloud migration:** Moving an application from AWS to Azure requires portable configurations and containerized deployment. - **Hybrid Cloud Operations:** Enterprises often integrate on-premise systems with cloud services, requiring APIs and standards that support interoperability. - **Vendor Neutral Tools:** Use of Docker, Kubernetes, and Terraform ensures applications can run on multiple cloud platforms without rewriting code.

Q2: Answer the following questions (Any THREE) - [5 Marks Each]

1. Differentiate between Vertical Scaling & Horizontal Scaling.

Feature	Vertical Scaling	Horizontal Scaling
Definition	Increasing resources in a single machine	Adding more machines/nodes
Implementation	Upgrade RAM/CPU	Add more servers to the pool
Downtime	Often requires downtime	Minimal or no downtime
Cost	Expensive at scale	Cost-effective for large systems
Example	Upgrade from 4GB to 16GB RAM	Add 3 web servers behind a load balancer

Vertical scaling is easier initially, while horizontal scaling is preferred for high availability and large-scale distributed systems.

2. Discuss classification of load balancing algorithms. Load balancing algorithms are used to distribute incoming network traffic across multiple servers: - **Static Algorithms:** - *Round Robin:* Requests are distributed sequentially. - *Weighted Round Robin:* Each server is assigned a weight; higher-weight servers handle more requests. - **Dynamic Algorithms:** - *Least Connection:* Routes traffic to the server with the fewest active connections. - *Resource Based:* Considers CPU, RAM usage before allocating requests. - **Content-Based Routing (Layer 7):** - Routes traffic based on content (e.g., URL paths).

Choosing the right algorithm improves availability and performance.

3. Discuss the model of CDN in detail. A **Content Delivery Network (CDN)** is a system of distributed servers that deliver content based on users' geographic locations. - **Components:** - *Origin Server:* Hosts the original version of content. - *Edge Servers:* Cache content closer to users. - *DNS & Routing Logic:* Directs user requests to the nearest edge server. - **Benefits:** - Reduces latency and server load. - Increases download speed. - Enhances scalability and availability. - Provides DDoS protection. - **Examples:** Cloudflare, AWS CloudFront, Akamai.

Q3: Answer the following in brief (Any THREE) - [2 Marks Each]

1. What role does the service level agreement (SLA) play in capacity planning? SLA defines the expected service performance (e.g., uptime, latency). It guides: - How much capacity to provision. - Ensures systems meet agreed performance targets. - Reduces business risks and customer dissatisfaction.

2. How persistence is implemented in load balancing? Persistence (also known as sticky sessions) ensures a client is always directed to the same server during a session. Techniques: - **IP Hashing** - **Session Cookie-based Routing** Useful for applications where session data is stored locally (e.g., login sessions).

3. What is the need for NoSQL database in cloud computing system? - Handles large volumes of unstructured/semi-structured data. - Supports horizontal scaling and distributed architecture. - Schema flexibility helps in agile development. - Examples: MongoDB, Cassandra, DynamoDB.

2 Sessional - 1

Q1: Answer the following in brief (Any THREE) - [3 Marks Each]

1. What do you understand by ‘Cloud’ in Cloud Computing? The term “Cloud” in Cloud Computing refers to on-demand availability of computing resources (like servers, storage, databases, networking, software) over the internet. These resources are managed by cloud providers and accessed remotely. It enables users to use IT infrastructure without owning or managing physical hardware.

2. Discuss how Cloud Computing can be beneficial to University. Cloud computing benefits universities by: - **Cost savings:** Pay-as-you-go model reduces hardware/software investment. - **Scalability:** Easily scale resources for students, exams, research. - **Collaboration:** Google Workspace, Microsoft 365 for sharing and group projects. - **Remote access:** Students and faculty can access resources anywhere. - **Data storage & security:** Centralized, secured data backup and access.

3. Discuss any 3 services of Cloud Computing with a case study of real-time scenario and its application. - **IaaS (Infrastructure as a Service):** AWS EC2 allows launching virtual machines. - **PaaS (Platform as a Service):** Google App Engine allows deploying applications without managing servers. - **SaaS (Software as a Service):** Google Docs enables collaborative document editing. *Case Study:* A university uses AWS EC2 for virtual labs and Google Docs for collaborative student assignments.

Q2: Answer the following questions (Any THREE) - [5 Marks Each]

1. Explain briefly Service Delivery model. Briefly explain any 4 different AWS cloud services. Cloud Delivery Models: - **IaaS:** Provides virtualized infrastructure (e.g., EC2). - **PaaS:** Platform for developing, testing applications (e.g., Elastic Beanstalk). - **SaaS:** Fully managed software (e.g., Amazon WorkDocs).

4 AWS Services: - **Amazon EC2 (IaaS):** Launch and manage virtual servers. - **Amazon S3:** Object storage service. - **Amazon RDS:** Managed database service. - **AWS Lambda:** Serverless compute service.

2. What do you understand by Cloud Deployment models? Discuss each model briefly. - **Public Cloud:** Services are delivered over the internet (e.g., AWS, Azure). - **Private Cloud:** Used exclusively by a single organization (e.g., government). - **Hybrid Cloud:** Combination of public and private clouds. - **Community Cloud:** Shared by organizations with similar needs (e.g., education, healthcare).

3. State the actors and their roles in Cloud model. - **Cloud Provider:** Offers services (e.g., AWS, Azure). - **Cloud Consumer:** Uses the services. - **Cloud Auditor:** Ensures compliance and security. - **Cloud Broker:** Manages service usage across multiple providers. - **Cloud Carrier:** Intermediary that provides connectivity and transport.

Q3: Answer the following in brief (Any THREE) - [2 Marks Each]

1. What activities can be performed by consumers of PaaS facility? - Develop, test, and deploy web apps. - Manage application lifecycle. - Use built-in development tools and APIs. - Focus on code without managing infrastructure.

2. What is the importance of the security reference model in cloud computing? - Provides structured approach for security architecture. - Identifies roles, policies, mechanisms. - Helps enforce access control, data encryption, and compliance.

3. Give two differences between Grid Computing & Cloud Computing. | Grid Computing | Cloud Computing | |-----|-----| | Resource-sharing across locations | On-demand service over internet | | Less scalable and flexible | Highly scalable and virtualized |

4. State the main 2 differences between Type-1 and Type-2 hypervisor. | Type-1 Hypervisor | Type-2 Hypervisor | |-----|-----| | Runs directly on hardware | Runs on host operating system | | Used in data centers | Used for desktop virtualization |

5. Why is cloud computing called variable cost computing model? Cloud services follow pay-per-use model: - No upfront capital expenses. - Cost depends on usage (compute hours, storage, bandwidth). - Allows businesses to control IT spending based on demand.
